Neuralink Digital Design Technical Challenge

RTL Focus

Introduction

The goal of this assignment is to design the microarchitecture of a message extractor for a 10G Ethernet processing pipeline and describe its operation.

Specification

The input to the pipeline is a 64b wide AXI stream slave interface, and the output is a 64b wide AXI stream master interface. The input stream contains a sequence of packets, each of which contains a number of variable length messages, as shown in Figure 1.

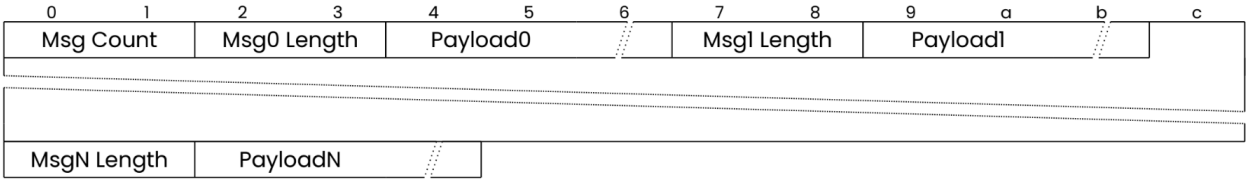

Figure 1. Incoming Packet format.

The expected output is the variable length payload extracted from each message. The input and output have the following specifications:

1. The input is a 64b wide AXI streaming slave interface.
2. The output is a 64b wide AXI streaming master interface.
3. The minimum length for any message is 8 bytes.
4. The maximum length for any message is 32 bytes.
5. The total length of a packet will not exceed 1500 bytes.
6. tready can be assumed to be always asserted for the output stream, i.e., there is no back pressure asserted into this module.
7. The input data is in network order, i.e., big endian.
8. The tready signal can be asserted combinationally to reduce pipeline bubbles, i.e., in the same cycle as tvalid being asserted.

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Example Waveforms

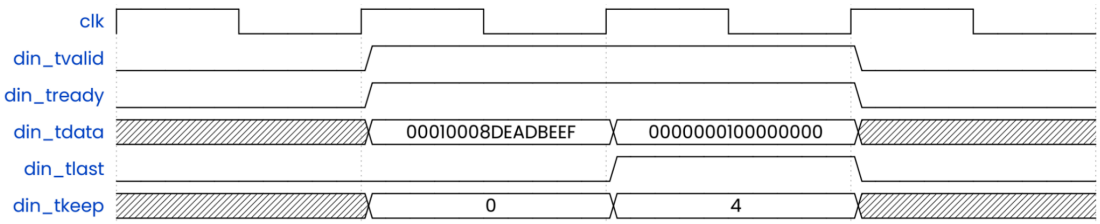

Figure 2. Input diagram for a packet containing a single 8 byte message.

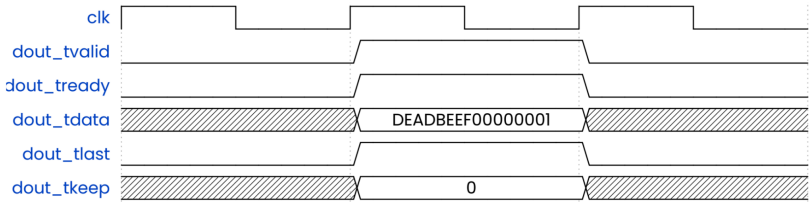

Figure 3. Possible output corresponding to the input in Figure 2.

In this first example (Figure 2 and Figure 3), a single 8 byte message with the contents `0xDEADBEEF00000001` is contained in the packet. The output is simply the 8 byte payload.

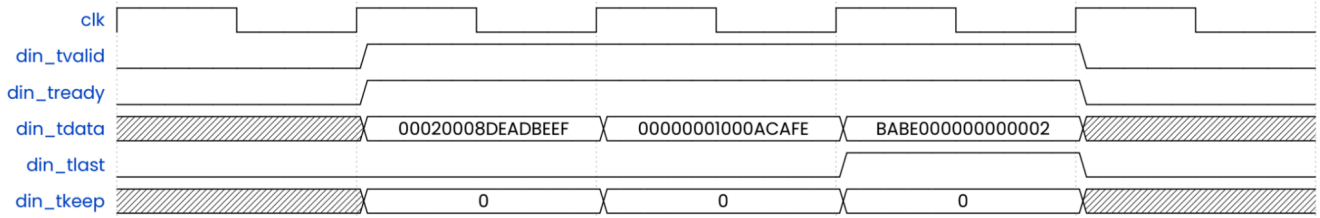

Figure 4. Input diagram for a packet containing two messages.

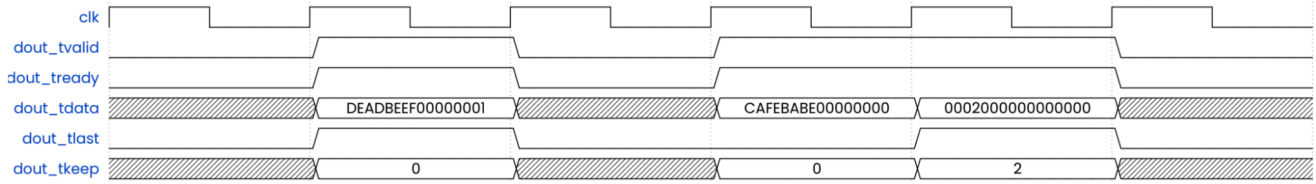

Figure 5. Possible output corresponding to the input in Figure 4.

In this second example (Figure 4 and Figure 5), two messages are included in the packet. The first has an 8 byte payload of `0xDEADBEEF0000001` and the second has a 10 byte payload of `0xCAFEBABE000000000002`. The output includes the two payloads, each terminated with `tlast`.

Tasks

1. Draw **block diagram(s) on the microarchitecture** of the proposed design, which can minimize the pipeline bubbles, and explain key data structures and reasoning behind your design decisions.
2. Draw **finite state machine(s)**, which describe the operation of the microarchitecture that you described, and present an implementation in synthesizable SystemVerilog.
3. Describe noteworthy **trade-offs in performance, power, and area** with your SystemVerilog implementation.
4. Present a **verification plan** for your SystemVerilog implementation, which describes workload and pass/fail criteria.
5. Write a **testbench** to verify the given design as planned above, and attach the verification testbench output.

Please describe any assumptions you have made. Feel free to reach out over email with any questions.